Simulation Homework

I am going to ask you to solve and simulate two systems using Matlab. You can either use the Matlab script or Simulink, it is your choice. There is one first order and one second order system. Solve the equation and simulate the system. Use a step input with an amplitude of 1 and choose your one time range.

Question 1 – Second order system

Regard the following differential equation:

$$2\ddot{x} + 4\dot{x} + 2x = 2y$$

1. Is this system Under damped, Critically damped or Heavily damped.
2. Give me solve differential equation.
3. Simulate the system with Matlab and give me a picture of the output.
4. Show me your code or Simulink model that you made.

Question 2 – First order system

Regards the following differential equation:

$$4\dot{x} + 1x = 0.5y$$

1. Give me solve differential equation.
2. Simulate the system with Matlab and give me a picture of the output.
3. Show me your code or Simulink model that you made.

Please answer question 1 and 2. Email me the pdf to Alfred.barnard@srh.de

Remember to include your name and student number in the document.